

MUHAMMED NURUDEEN OLALEKAN

Python Backend Engineer | AI/ML Engineer

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PROFESSIONAL SUMMARY

Python Backend and AI/ML Engineer with a B.Sc. Mathematics background, building REST APIs in FastAPI and Django and machine learning systems in PyTorch and scikit-learn. Delivered engineering work across agentic LLM applications (LangChain, LangGraph, MCP tool calling, Ollama-hosted models), computer-vision segmentation pipelines, and modular data engineering workflows in pandas. Hands-on with MySQL and PostgreSQL schema design, AWS EC2, Docker Compose, Linux and pytest. Teaches Python web development and AI/ML to early-career developers.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript

Backend and APIs: FastAPI, Django, REST API design, API versioning, Pydantic, SQLAlchemy, JWT authentication, role-based access control

Databases: MySQL, PostgreSQL, schema design, query optimisation

AI and Machine Learning: PyTorch, scikit-learn, scikit-image, OpenCV, Albumentations, LLM integration, LangChain, LangGraph, Model Context Protocol (MCP), tool calling, Ollama, NLP, predictive analytics, anomaly detection

Data and Analytics: Pandas, NumPy, matplotlib, seaborn, Power BI, Excel, ETL, data wrangling

Cloud and DevOps: AWS (EC2), Docker, Docker Compose, Linux/Ubuntu, Git, GitHub, Vercel, Netlify

Testing: pytest, unit testing

Frontend: React.js, Next.js, HTML5, CSS3, TailwindCSS

PROFESSIONAL EXPERIENCE

Software Engineer and Python Instructor | Sysbeams | Abeokuta, Nigeria | Jan 2026 - Present

Engineered the backend of an Incident Report Management System using FastAPI and MySQL, implementing role-based access control and JWT authentication to restrict data access by user tier.

Integrated Google's Gemini LLM to produce structured incident analysis, automating report summarisation previously handled manually.

Built an agentic AI assistant for the university portal in FastAPI, calling the school's existing ASP.NET Core services and passing the user's JWT through to the backend so the model acts on their behalf.

Refactored the assistant onto a Model Context Protocol (MCP) architecture, granting the model scoped tool-level access to the database in place of a fixed endpoint contract.

Streamed Ollama-hosted model responses to the client over asynchronous HTTP, with Next.js session state preserved across page navigation.

Developed the React.js frontend for the university portal, building dynamic dashboards for academic operations, user management and administration.

Designed and delivered a structured curriculum in Python web development and AI/ML covering fundamentals, data structures, OOP and software engineering practice for 20+ students.

Mentored junior developers on REST API design patterns, clean code principles and Git workflows.

AI Engineer (Contract) | Jupiter AI Labs | Remote | Dec 2025 - Present

Evaluated LLM-generated responses against defined quality and accuracy criteria, providing structured feedback to guide improvements in model output.

Assessed responses for correctness, relevance and instruction adherence across use cases to support model quality assurance.

Freelance Python, Data and AI Engineer | Remote | Apr 2025 - Present

Designed and delivered 10+ REST APIs in FastAPI and Django with JWT authentication for client engagements.

Built ETL pipelines in Pandas and NumPy, ingesting, cleaning and transforming datasets for client reporting and Power BI dashboards.

Developed machine learning models for predictive analytics and NLP tasks including text classification and sentiment analysis, integrated into production web applications.

Integrated OpenAI LLM APIs into client workflows to automate content generation, customer support and data summarisation.

Deployed services to AWS EC2 and ran application components as containers using Docker Compose.

Optimised relational schemas and queries in PostgreSQL and MySQL to improve data integrity and retrieval performance.

PROJECTS

Agentic University Portal Assistant - Python, FastAPI, MCP, Ollama (Qwen 2.5), Next.js

github.com/GATEWAY0710/student-support-ai-chatbox

Delivered an LLM agent that answers student queries on results, timetables and fee status in natural language, deployed as a chat interface in the university portal.

Served as a stateless proxy: the frontend passes the user's JWT to FastAPI, which forwards it to the school's backend so the model never manages its own authentication state.

Migrated the integration to MCP, replacing fixed endpoint calls with tool-level database access scoped to the authenticated user.

Garment Segmentation Pipeline - Python, PyTorch, Albumentations, OpenCV, scikit-image, pytest

github.com/GATEWAY0710/ML-ENGINEER-ASSESSMENT

Built a training and inference pipeline for garment segmentation, with separate training, prediction and post-processing scripts and versioned model weights.

Applied Albumentations for data augmentation and OpenCV for mask post-processing, tracking configurations through YAML files.

Crypto Market Data Engineering Pipeline - Python, Pandas, NumPy, matplotlib, seaborn

github.com/GATEWAY0710/Muhammed_Nurudeen_-ARK-Data-engineer-Take-home

Built a modular pipeline for a data engineering take-home: data loading, signal generation, a backtest engine, performance metrics, visualisation and export.

Processed hourly and 4-hourly BTC/USDT kline datasets, separating ingestion from analysis into independently testable modules.

Gateway Store (E-Commerce Application) - Python, FastAPI, Django, SQLAlchemy

github.com/GATEWAY0710/GatewayStoreFastApi

Rebuilt an initial Django implementation as a FastAPI service, structuring data access through SQLAlchemy models and RESTful endpoints.

EDUCATION

B.Sc. Mathematics | Federal University of Agriculture, Abeokuta, Nigeria | 2018 - 2023

Second Class Lower Division (2:2)

CERTIFICATIONS

AI Agents Fundamentals | Hugging Face | Jul 2026

AWS Scholars Program Project Badge: Analyze Data using AI with PartyRock | Udacity | Mar 2026

ADDITIONAL INFORMATION

Languages: English (Fluent), Yoruba (Native)

National Youth Service Corps: Teaching Assistant, 2024 - 2025

Availability: Open to full-time, contract and remote roles